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标题	Synthesis and Functional Properties of Fluorinated Phosphonate‐Substituted Porphyrins: Excellent Photosensitizers and Robust Precursors for Photoactive Langmuir – Schaefer Films
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摘要（原文节选）

ABSTRACTFluorinated porphyrins have attracted considerable attention in PDT, sensing and catalysis. Guided by demands for functional porphyrins with tunable electronic properties, high photostability, and controlled supramolecular organization, we report a synthetic approach to meso‐tetrakis[4‐(diethoxyphosphoryl)‐2,3,5,6‐tetrafluorophenyl]porphyrin (H2TF4PPP), which combines the advantages of diethyl phosphonate and fluorine substituents within a highly symmetrical porphyrin molecule. X‐ray diffraction analysis of ⋯

数据表 · TABLE 2（7 行，含数值行 7 行）

Singlet oxygen quantum yields of MTF4PPP and MTPPP. a

Compound	Solvent	φ Δ	Reference
H2TF4PPP	CHCl3	0.83	This work
H2TPPP	CHCl3	0.82	This work
	CD3CN/D2O (10:1)	0.72	[51]
PdTF4PPP	CHCl3	0.97	This work
PdTPPP	CHCl3	1.1	This work
	CD3CN/D2O (10:1)	0.94	[51]
ZnTF4PPP	CHCl3	0.9	This work

脚注（逐字）：aComparative studies by singlet oxygen phosphorescence measurements in chloroform using PH as a standard. Quantum yield values are given with 10% error.

正文对应句（逐字）：Fluorinated meso‐tetraarylporphyrins are typically synthesized using the nucleophilic substitution of 5,10,15,20‐tetrakis(pentafluorophenyl)porphyrin (H2TF5PP), which regioselectively reacts with N‐, S‐, and O‐nucleophiles to afford, in most cases, tetra‐substituted derivatives bearing the nucleophi

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